

Xiangjian Zeng

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Research Objective

To research brain physiology with computational modeling and create a functional neuromodulation system with engineering toolsets

Education

Massachusetts Institute of Technology, Cambridge, MA

Expected 06/2029

Ph.D in Electrical Engineering and Computer Science, GPA: 5.00/5.00

Carnegie Mellon University, Pittsburgh, PA

M.S. in Electrical and Computer Engineering, GPA: 4.00/4.00

12/2023

University of California San Diego, La Jolla, CA

B.S. in Electrical and Computer Engineering (Minor in Cognitive Science), GPA: 3.91/4.00

12/2020

Honor: Magna Cum Laude

Research Experience

Lewis Lab, Massachusetts Institute of Technology, Cambridge, MA

09/2024-present

Closed-loop Auditory Modulation of Cerebrospinal Fluid during Sleep

- Develop a closed-loop neuromodulation system that performs real-time scalp-EEG filtering and machine-learning-based slow-oscillation phase prediction to deliver precisely timed auditory stimulation during sleep
- Integrate simultaneous EEG-fMRI with electrophysiological and CSF-flow analyses to quantify stimulation-evoked brain and fluid dynamics across younger and older adults
- Co-authored an accepted Science Translational Medicine study demonstrating that closed-loop auditory stimulation targeting sleep slow waves modulates cerebrospinal fluid flow in humans, motivating development of personalized stimulation algorithms

Insanally Lab, University of Pittsburgh, Pittsburgh, PA

05/2023-08/2024

Bayesian Ensemble Neural Decoding

- Developed a probabilistic ensemble neural decoder, combining inter-spike-interval-based conditional probabilities across multiple neurons to jointly predict behavioral trial conditions, revealing that traditionally neglected “quiet” non-classically responsive neurons interact with responsive neurons in learning
- Demonstrated that aggregating non-classically responsive population activity improved behavioral decoding accuracy from 60% to 65% during late-stage learning compared to early, revealing information not captured by classical response-selective neurons

Spiking Recurrent Neural Network Simulation of the Brain

- Built spiking recurrent neural-network models with 1000 units in Python and Julia under biological constraints to simulate cortical mechanisms underlying auditory discrimination behavior
- Investigated how flexible neural-network dynamics emerge from balanced excitatory and inhibitory activity and authored a manuscript describing the resulting computational model

Chamanzar Lab, Carnegie Mellon University, Pittsburgh, PA

05/2022-04/2023

Neural Probe Waveguide Characterization System

- Developed an automated, non-destructive characterization system for photonic waveguides used in neural optical probes by integrating robotics, computer vision, and signal processing

- Demonstrated waveguide-loss characterization previously achievable primarily through destructive methods and first-authored a peer-reviewed conference proceedings paper describing the pipeline

Lerman Lab, UC San Diego, La Jolla, CA

03/2020-01/2021

Adaptive Neural Signal Noise Cancellation

- Led development of an adaptive filtering algorithm for suppressing power-line interference in peripheral nerve recordings and implemented the method in a hardware demonstration
- Achieved approximately 45 dB suppression of 60-Hz interference and presented the work as first author at the North American Neuromodulation Society meeting

Peripheral Nerve Magnetoneurography

- Developed MATLAB signal-cleaning and processing pipelines for magnetic measurements of peripheral nerve activity, contributing analyses to a collaborative publication
- Supported human neuroscience studies involving peripheral nerve measurements and electrical stimulation

Spitzer Lab, UC San Diego, La Jolla, CA

10/2019-12/2020

Automated Fluorescence Cell Counting

- Developed an automatic cell recognition program for fluorescence brain slices in Matlab utilizing correlation to count and map a variety of brain cells, obtaining an accuracy that's close to expert's manual selection

Neuroelectronics Group, La Jolla, San Diego, CA

07/2019-08/2019

Neural Recording Amplifier PCB Design

- Designed, assembled, and tested an Altium PCB for an amplifier used in in vivo neural recording
- Reduced the redesigned amplifier-board footprint to approximately 3 cm × 1 cm, more than twofold smaller than the previous design

Publications

Paper Publications

- Levitt, J., **Zeng, X.**, Yang, Z., Jacob, L. P. L., Lewis, L. D. Closed-loop auditory stimulation targeting sleep slow waves drives cerebrospinal fluid flow in humans. *Science Trans. Med.* (accepted).
- **Zeng, X.**, Reddy, J., Chamanzar, M. Automated Photonic Waveguide Loss Measurement using Out-scattering Light Method. *Integrated Systems Design and Technology Conference 2023 book proceedings.*
- Toth, J., Sidleck, B., Lombardi, O., Hou, T., Eldo, A., Kerlin, M., **Zeng, X.**, Saeed, D., Agarwal, P., Leonard, D., Andrino, L., Inbar, T., Insanally, M. Dynamic gating of perceptual flexibility by non-classically responsive neurons (manuscript in review).

Conference Publications

- **Zeng, X.**, Levitt, J., Lewis, L.D. Auditory Stimulation During Sleep Modulates CSF Flow via Dual Slow-Wave and High Frequency Pathways. *Society for Neuroscience, San Diego, USA, Nov 15-19, 2025.*
- **Zeng, X.**, Toth, J., Albana, B. F., Insanally, M. N. Contributions and synaptic basis of diverse cortical neuron response to flexible task performance. *Society for Neuroscience, Chicago IL, United States, Oct 5-9, 2024.*
- **Zeng, X.**, Yu, G., Albana, B. F., Insanally, M. N. Contributions and synaptic basis of diverse cortical neuron response to flexible task performance. *COSYNE, Montreal, Canada, Mar 27-30, 2024.*
- **Zeng, X.**, Lerman, I., Bu, Y., Kimball, D. RLS Harmonic Powerline Noise Cancellation in Radial Nerve Recording. *North American Neuromodulation Society Virtual Meeting, Orlando FL, United States, Jan 15-16, 2021.*
- Bu, Y., Lerman, I., Borna, A., Coleman, T., Rao, R., Huang, M., Shah, V., Kimball, D., Morton, T., **Zeng, X.** Capturing the Magnetic Signature of Small Fiber Neuronal Activity. *North American Neuromodulation Society Virtual Meeting, Orlando FL, United States, Jan 15-16, 2021.*

- Hou, T., Toth, J., Sidleck, B., Lombardi, O., Eldo, A., Saeed, D., Kerlin, M., **Zeng, X.**, Agarwal, P., Leonard, D., Andrino, L., Inbar, T., Insanally, M. N. Dynamic gating of perceptual flexibility by diverse cortical responses. Society for Neuroscience, 2023.

Fellowship Awards

- EECS MathWorks Fellowship** 2026
- MIT Artificial Intelligence and Wellness Fellowship** 2025

Academic Project

Deep Learning Automatic Fake News Detection Fall 2022

- Jointly created a BERT based language model to automatically extract relevant evidence and classify misinformation from political statements, achieving a two-way accuracy of 72%

EEG-based Music Recommendation Fall 2022

- Led to write a K Nearest Neighbor(KNN) prediction system to automatically detect musical preferences from brain EEG recordings of listening subjects and update music recommendation in real-time

Relevant Courses

Computing: CMU - Intro to Deep Learning, Machine Learning for Signal Processing

Neuroscience: MIT - Functional Magnetic Resonance Imaging: Data Acquisition and Analysis, CMU - Neuroengineering Laboratory, Advanced System Neuroscience, UCSD - Cognitive Neuroscience

Electrical Engineering: UCSD - Linear Electronic Systems, Introduction to Digital Design, Components and Circuits Laboratory

MEMS: CMU - Micro and Nano Systems Fabrication, Nano-Bio-Photonics

Technical Skills

Programming & ML: Python, C/C++, MATLAB, Julia, R | PyTorch, TensorFlow, scikit-learn

Neural Data & Modeling: EEG, EEG-fMRI, electrophysiology, neural decoding, adaptive filtering, computational modeling

Neurotechnology: Closed-loop stimulation, peripheral nerve recording, real-time signal processing

Hardware & Prototyping: PCB design, Altium, microfabrication, Simulink, SolidWorks

References

Professor Laura D. Lewis, Electrical Engineering and Computer Science, Massachusetts Institute of Technology, ldlewis@mit.edu

Professor Michele Insanally, Otolaryngology & Neurobiology, University of Pittsburgh
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Professor Badr F. Albanna, Neuroscience, University of Pittsburgh School of Medicine
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